

RiskShield-PPO: Safe Reinforcement Learning for Construction-Site Inspection under Perception Uncertainty, with an Experimental Hierarchical Extension

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Index Terms—safe reinforcement learning, construction robotics, predictive safety shielding, autonomous inspection, perception uncertainty, Webots

Abstract—Autonomous construction inspection couples sparse coverage objectives with collisions, worker proximity, restricted areas, and imperfect visual perception. We present a reproducible Gymnasium benchmark, RiskShield-PPO (PPO-Lagrangian with an auxiliary cost-value estimator), and a predictive action shield. A five-algorithm matrix contains 1,350 episodes over three site sizes, three hazard densities, three perception-noise levels, and ten paired seeds; six ablations add 180 episodes. RiskShield-PPO reduced mean safety cost from 49.3852 for PPO to 17.1296 and reduced collision rate from 0.0741 to zero, but both learned methods had zero complete-mission success. Frontier exploration reached 0.9667 success and 0.9978 hazard recall, while risk-aware A* reached 1.0 for both metrics. A separate 320-episode uncertainty suite across 64 conditions had mean robustness 0.8893. We additionally report an experimental hierarchical v4 protocol over 270 held-out episodes, with 0.5889 mission success and 0.9214 hazard recall; v4 does not replace production RiskShield-PPO v1. The evidence is simulation-only and does not establish real-world safety.

I. INTRODUCTION

Construction inspection requires visiting hazards while avoiding workers, restricted regions, machinery, and fall risks. Reward maximization alone does not express an acceptable risk budget, and perception errors make fixed safety assumptions unreliable. Safe RL commonly modifies the objective or constrains exploration [1]; constrained policy optimization formalizes cost limits [2], while shields can replace actions before execution [3].

The contributions are: (1) a typed, deterministic construction-inspection benchmark with expert, PPO, SAC, and constrained-policy baselines; (2) RiskShield-PPO, which separates the learned cost signal from a predictive runtime shield; (3) a controlled perception-uncertainty protocol; (4) an integrated Webots execution path with traceable shield-to-motor commands; and (5) an experimental hierarchical extension with persistent targets and local planning. We make no claim of learned-policy task mastery: the complete primary matrix shows zero mission success for all three learned policies.

II. RELATED WORK

PPO optimizes a clipped policy-ratio surrogate [4]. SAC is an off-policy maximum-entropy actor-critic method [5]. CMDP methods optimize reward subject to expected cost [2]. Shielding instead filters proposed actions with an external safety model [3]. Our system combines a Lagrangian training signal with a predictive runtime filter; it does not inherit formal guarantees from either cited method. Gymnasium provides the standardized environment API [6], and Webots supplies the robot simulator [7].

III. PROBLEM FORMULATION

We model a finite-horizon CMDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r, c, \gamma, d)$. The discrete actions are north, south, west, east, and inspect. The observation contains padded semantic map channels (obstacle, hazard, worker, restricted, visited, agent, risk, dynamic hazard, PPE risk, visibility), pose, orientation, progress, local risk, and an action mask.

Reward separates task progress from cost:

$$r_t = -0.01 + r_{\text{inspect}} + r_{\text{coverage}} - r_{\text{invalid}}, \quad (1)$$

while

$$c_t = c_{\text{collision}} + c_{\text{near}} + c_{\text{restricted}} + c_{\text{PPE}}. \quad (2)$$

The objective is

$$\max_{\pi} J_R(\pi) \quad \text{s.t.} \quad J_C(\pi) \leq d, \quad (3)$$

with training budget $d = 5$. Reward and cost remain separately logged.

IV. CONSTRUCTION SAFETY RL BENCHMARK

Three parameterized grid sites vary size, hazards, workers, dynamic obstacles, and episode horizon. Reset and transition randomness use explicit seeds. Obstacle-aware A*, frontier exploration, and nearest-risk revisit provide interpretable expert components. The final comparison uses risk-aware A*, frontier exploration, PPO, SAC, and RiskShield-PPO.

The environment includes partial observation, semantic layers, action masks, dynamic hazards, restricted zones, PPE-risk cells, inspection targets, reward, cost, rendering, and serialized telemetry. The Gymnasium checker, deterministic reset/step, mask, collision, near-miss, cost, reward, and serialization tests pass.

TABLE I

PROTOCOL-SPECIFIC HIERARCHICAL-v4 EVIDENCE. DASHES INDICATE METRICS NOT RECORDED BY THE CORRESPONDING CANONICAL REPORT; ROWS ARE NOT DIRECTLY COMPARABLE BECAUSE TRAINING STEPS, CHECKPOINTS, AND EVALUATION PROTOCOLS DIFFER.

Variant / protocol	Success	Recall	Cost	Evidence scale
H3 preliminary	0.6154	0.9551	20.3654	5k option steps; 76 episodes
H4 imitation	0.6250	0.9225	16.3125	paired eval; n unreported
H4 recovery 30k	0.5625	–	17.1406	30k option steps
Final held-out v4	0.5889	0.9214	0.0667	270 episodes

V. RISKSHIELD-PPO METHOD

RiskShield-PPO uses PPO’s clipped objective but trains under a Lagrangian penalty:

$$\mathcal{L}(\theta, \lambda) = J_R(\pi_\theta) - \lambda (J_C(\pi_\theta) - d), \quad \lambda \geq 0. \quad (4)$$

The multiplier update is

$$\lambda_{k+1} = \Pi_{[0,100]} [\lambda_k + \eta_\lambda (\hat{J}_C - d)], \quad (5)$$

with $\eta_\lambda = 0.02$. An auxiliary neural critic minimizes

$$\mathcal{L}_C = (V_C(s_t) - [c_t + \gamma(1 - z_t)V_C(s_{t+1})])^2. \quad (6)$$

The resulting penalized reward actually reaches PPO training. Telemetry records policy/value diagnostics, cost-value loss, cost, budget, and multiplier. Training stopped after 15,300 steps without improvement; the multiplier saturated, a negative result reported rather than hidden.

VI. HIERARCHICAL EXPERIMENTAL SYSTEM

The experimental v4 system separates learned mission decisions from local execution. A recurrent option policy selects exploration, inspection, retreat, and replanning intentions and a target in the causal observed map. An incremental risk-aware planner proposes a local route; a predictive controller and Safety Shield validate the resulting primitive before wheel commands. The policy never receives hidden map state or future worker motion. This is not end-to-end wheel-velocity reinforcement learning. The protocol-specific results and their denominators are reported in Table I. These results motivate retaining v1 in production.

VII. PREDICTIVE SAFETY SHIELD

For proposed action a_t , the shield rolls forward a deterministic local model for horizon k . It predicts collisions, near misses, restricted entry, proximity, and budget excess. It accepts, replaces, masks, or stops the action.

The structured record includes proposed and final actions, trajectory, cost, violations, decision, confidence, emergency state, and latency. In Webots, the final shielded action, not the proposal, reaches wheel commands.

Algorithm 1 Predictive shield

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1: Receive  $s_t$ , proposed  $a_t$ , budget  $b_t$ 
2: for  $a$  in proposed-first candidate order do
3:    $\tau, \hat{C} \leftarrow \text{Rollout}(s_t, a, k)$ 
4:   if  $\tau$  has no violation and  $\hat{C} \leq b_t$  then
5:     return structured decision with final action  $a$ 
6:   end if
7: end for
8: return emergency stop with “no safe action”

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VIII. PERCEPTION AND SEMANTIC RISK MAPPING

The integrated path is Webots camera $\rightarrow$ CUDA detector $\rightarrow$ typed detections $\rightarrow$ temporal filtering $\rightarrow$ semantic risk state $\rightarrow$ policy and shield. The production checkpoint has 14 declared classes and a recorded SHA-256 hash. Unsupported concepts use explicitly labeled simulator ground truth rather than fabricated detections. The corrected hierarchical Webots runtime records 100 policy decisions per world with real camera timestamps, recurrent-state hashes, target persistence, planner/controller/shield traces, and wheel commands. In the tested worlds the detector produced no valid detections, so the v4 report truthfully leaves perception-driven semantic change false; no CV metric is fabricated.

IX. EXPERIMENTAL SETUP

The primary matrix is 3 site sizes $\times$ 3 hazard densities $\times$ 3 false-negative probabilities $\times$ 5 algorithms $\times$ 10 paired evaluation seeds, totaling 1,350 episodes. Cache records include immutable configuration and checkpoint hashes. Planner episodes run concurrently; GPU policy inference is serialized. No missing run is imputed.

We report mean, standard deviation, median, IQR, and 95% t intervals per cell. Friedman tests compare all algorithms because bounded, zero-inflated metrics need not be normal. A paired Wilcoxon comparison contrasts RiskShield-PPO and PPO, with Holm correction over six metrics. Standardized paired mean difference is the effect size.

The corrected Webots integration was validated headlessly in the small, medium, and dynamic worlds. Each run reached 100 hierarchical decisions and used the H4 target-persistence checkpoint, while RiskShield-PPO v1 remained the production policy. The earlier three-step adapter is retained only as superseded startup evidence. A visible real-time demo was attempted with the same controller; desktop Qt startup is environment-dependent and is reported separately from the accepted headless runtime evidence. Final user-facing visual demonstrations use a site-centered overview view so robot motion, obstacles, workers, and construction structures remain visible together. The final interactive Windows/Webots demonstration was manually run and visually verified using that overview configuration; this is separate from the environment-limited non-interactive Qt smoke.

The complete paired statistical analysis is summarized in Table II. The Friedman test is omnibus across the five algorithms; the paired Wilcoxon comparison is RiskShield-PPO

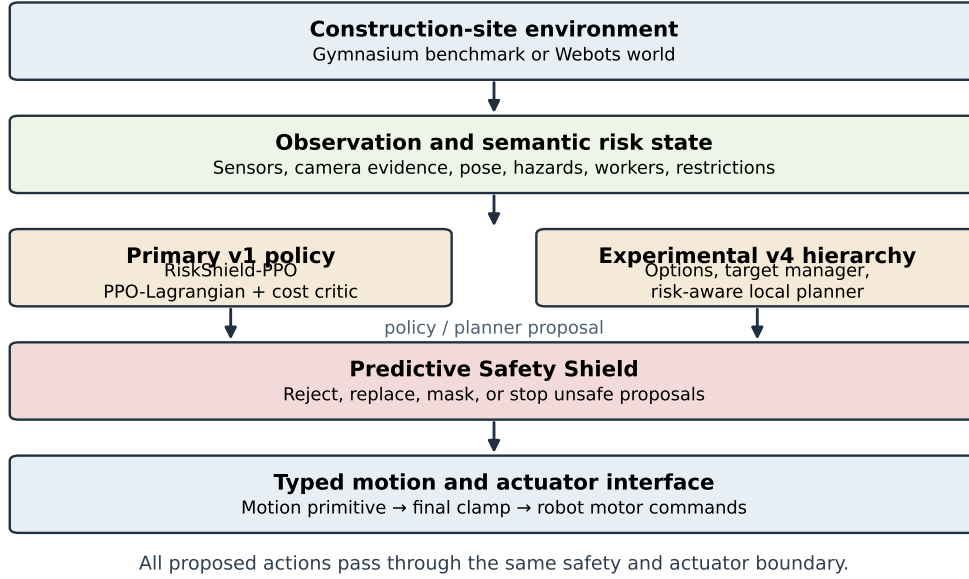


Fig. 1. Runtime architecture derived from the implemented benchmark and Webots paths. The primary RiskShield-PPO v1 policy and the experimental hierarchical v4 option/planner path both pass proposed actions through the same predictive Safety Shield and typed actuator boundary.

versus PPO, with Holm correction across the six planned metrics. These tests describe the primary benchmark only and do not turn the experimental v4 rows into a production claim.

X. MAIN RESULTS

RiskShield-PPO improved PPO’s hazard recall (0.147 versus 0.083), coverage (0.0275 versus 0.0090), and safety cost (17.13 versus 49.39), with no observed collision. SAC reached 0.074 coverage but incurred 150.86 mean safety cost and a 0.796 collision rate. Neither learned policy completed a mission. Frontier exploration achieved 0.967 success and 0.998 recall; risk-aware A* reached 1.0 success and recall but lower coverage.

XI. UNCERTAINTY EXPERIMENTS

The dedicated uncertainty suite evaluates false-negative probabilities $\{0, 0.1, 0.2, 0.3\}$ under four environment perturbation levels and four visual conditions, with five seeds per condition (320 episodes across 64 conditions). Clean truth, perturbed perception, and observation are kept separate. The canonical summary reports mean robustness 0.8893, with zero mission success in this learned-policy protocol. Figure 3 shows the measured hazard recall by environment perturbation; it is distinct from the primary benchmark’s perception-noise aggregation.

XII. ABLATION STUDY

Removing the predictive shield raised mean safety cost from 8.43 to 66.52. The one-step shield produced lower cost (2.07) than $k = 3$ in this limited sample, so longer prediction was not

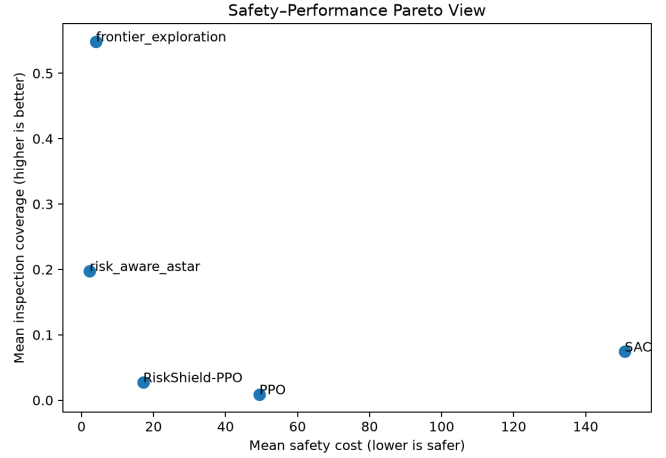


Fig. 2. Safety cost versus inspection coverage from raw primary records.

uniformly better. Removing CV risk injection or replacing the constrained checkpoint with PPO yielded identical reported task means under the shield, although their internal actions and intervention counts differed. This indicates that the shield dominated these weak policies. Removing action masking reduced coverage.

XIII. DISCUSSION

The planners’ advantage indicates that exploration and long-horizon credit assignment, not only collision avoidance, remain unsolved. RiskShield-PPO’s lower cost is meaningful but

TABLE II
PRIMARY BENCHMARK STATISTICAL RESULTS ($n = 270$ PAIRED EPISODES PER METRIC). EFFECT SIZE IS THE STANDARDIZED PAIRED MEAN DIFFERENCE FOR RISKSHIELD-PPO MINUS PPO.

Metric	Friedman statistic	Friedman p	Holm-adjusted Wilcoxon p	Effect size
Hazard recall	959.1985	2.48×10^{-206}	2.95×10^{-16}	0.5533
Inspection coverage	977.0342	3.38×10^{-210}	3.13×10^{-28}	0.7716
Collision rate	783.7004	2.61×10^{-168}	1.55×10^{-5}	-0.2823
Constraint violations	488.3545	2.21×10^{-104}	8.43×10^{-14}	-0.3444
Safety cost	622.1118	2.54×10^{-133}	8.31×10^{-18}	-0.4183
Success rate	1056.5393	1.99×10^{-227}	1.0000	0.0000

TABLE III
PRIMARY BENCHMARK MEANS OVER 270 EPISODES PER ALGORITHM.

Algorithm	Runs	Recall	Coverage	Collision rate	Safety cost	Success	Robustness
PPO	270	0.0834	0.0090	0.0741	49.3852	0.0000	0.0962
RiskShield-PPO	270	0.1466	0.0275	0.0000	17.1296	0.0000	0.1624
SAC	270	0.1339	0.0744	0.7963	150.8556	0.0000	0.0974
Frontier exploration	270	0.9978	0.5487	0.0000	4.0111	0.9667	0.6935
Risk-aware A*	270	1.0000	0.1976	0.0000	2.2500	1.0000	0.6085

TABLE IV
SIX PAIRED ABLATIONS; 30 EPISODES PER CONFIGURATION.

Variant	Episodes	Recall	Coverage	Safety cost	Collision rate	Success
One-step shield	30	0.1661	0.0569	2.0667	0.0000	0.0000
Full RiskShield-PPO	30	0.1522	0.0408	8.4333	0.0000	0.0000
Without action masking	30	0.1211	0.0206	5.9000	0.0000	0.0000
Without cost objective	30	0.1522	0.0408	8.4333	0.0000	0.0000
Without CV risk	30	0.1522	0.0408	8.4333	0.0000	0.0000
Without predictive shield	30	0.1189	0.0291	66.5167	0.0000	0.0000

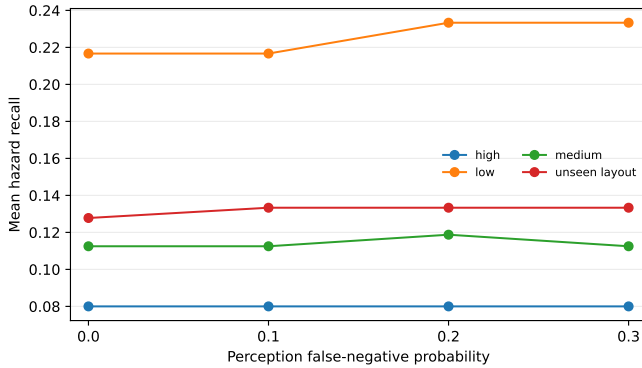


Fig. 3. Hazard recall in the dedicated 320-episode uncertainty suite. Each curve averages the four visual conditions at each measured false-negative probability; source data are in `reports/final_submission/stage8_uncertainty`.

insufficient: a stationary or repeatedly corrected policy can be safe and useless. The saturated multiplier suggests the selected budget was difficult relative to the training distribution. Future

work should increase training budgets, tune cost scaling, evaluate recurrent policies only after training a valid checkpoint, and separate shield dependence from intrinsic policy safety.

XIV. LIMITATIONS

All research evaluation is simulation-only. Learned policies had zero complete mission success. Stage 5C is a bounded integrated demonstration, not a long-duration deployment. The detector supports only its recorded class list; other semantic layers use simulator truth. No recurrent checkpoint exists, so no recurrent ablation is claimed. Results cover one laptop GPU and Webots R2025a. The shield is model-based and can fail under unmodeled dynamics. No real-world safety guarantee, certification, or human-subject evaluation is provided.

XV. ETHICAL AND SAFETY CONSIDERATIONS

The system must not replace qualified safety personnel or authorize entry into hazardous areas. Simulator performance can create false confidence. Deployment would require site-specific risk assessment, fail-safe hardware, independent verification, privacy controls, worker consultation, and compliance with local law and safety standards.

XVI. CONCLUSION

This work provides a reproducible construction Safe RL benchmark and truthful evidence spanning planning, learned policies, constrained optimization, shielding, uncertainty, CUDA perception, and Webots motor control. RiskShield-PPO reduced cost but did not solve inspection; expert planners remained substantially stronger. The hierarchical v4 extension demonstrates an additional target-and-planning protocol but does not replace v1. The negative result defines a clear research target rather than a deployment claim, and no real-world safety guarantee is established.

REPRODUCIBILITY STATEMENT

The repository records configuration files, seeds, checkpoint hashes, atomic per-run caches, raw CSV and Parquet data, statistical scripts, figure source data, tests, Webots R2025a metadata, and PowerShell entry points. The Python dependency contract is declared in `pyproject.toml`; the validated runtime uses Python 3.11. The complete benchmark command resumes only missing runs. The production command separately reproduces the Webots integration evidence. Large local checkpoints, datasets, and generated runtime workspaces are intentionally not mirrored in the public Git tree, so workflows requiring those artifacts must obtain them from the documented local manifests.

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